SCR2043 OPERATING SYSTEMS

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Section	: 02	

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to process monitoring and management in operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment02-studentname-matricno.docx**.

Essential Steps Before Starting Lab Assessment 2:

1. **Download necessary source codes:**

   Use the `wget` command to retrieve the following source code files to your Linux (or WSL or MacOS) environment:

```
wget -O mainprocess.c https://rebrand.ly/mainprocess_c
wget -O subprocess1.c https://rebrand.ly/subprocess1_c
wget -O subprocess2.c https://rebrand.ly/subprocess2_c
```

2. **Compile the source files:**

   Use the `gcc` compiler to create executable files from the source code.

```
gcc mainprocess.c -o mainprocess
gcc subprocess1.c -o subprocess1
gcc subprocess2.c -o subprocess2
```

3. **Execute the dummy processes:**

   Run all the dummy processes

```
./mainprocess &
```

   Press **enter** two times.

4. The dummy processes are running for 2 hours. If you took longer than 2 hours on questions 1-9, please restart the main process with `./mainprocess &`.

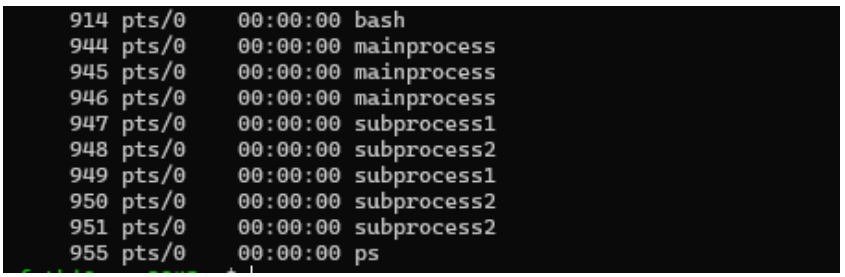
Lab Assessment 2 : Linux Process Monitoring and Management

Instructions:

1. Carefully execute each command as instructed in the questions.
2. Write down the exact command used for each task.
3. Capture a screenshot of the command's output.

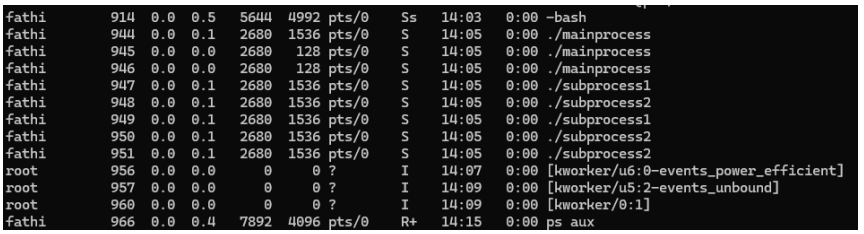
Question 1

Use the `ps` command with the appropriate option to display a complete list of all running processes within the Linux operating system.

Command
`ps -e`
Output


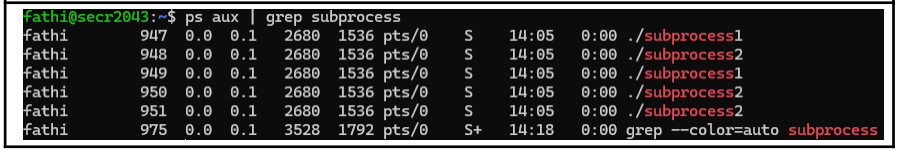
Question 2

Employ the `ps` command with necessary options to unveil comprehensive details about each running process.

Command
`ps aux`
Output


Question 3

Use the `ps` command with some tools to only list processes named "subprocess" and show some info about them.

Command
`ps aux | grep subprocess`
Output


Question 4

Execute the `ps` command, specifying options that reveal only the following columns:

- Process ID (`pid`)
- Owner of the process (`user`)
- CPU percentage (`pcpu`)
- Memory percentage (`pmem`)
- Command (`cmd`)

Command

```
ps -eo pid,user,pcpu,pmem,cmd
```

Output

```
  914 fathi     0.0  0.5 -bash
  944 fathi     0.0  0.1 ./mainprocess
  945 fathi     0.0  0.0 ./mainprocess
  946 fathi     0.0  0.0 ./mainprocess
  947 fathi     0.0  0.1 ./subprocess1
  948 fathi     0.0  0.1 ./subprocess2
  949 fathi     0.0  0.1 ./subprocess1
  950 fathi     0.0  0.1 ./subprocess2
  951 fathi     0.0  0.1 ./subprocess2
  956 root      0.0  0.0 [kworker/u6:0-events_power_efficient]
  957 root      0.0  0.0 [kworker/u5:2-events_power_efficient]
  967 root      0.1  0.0 [kworker/0:2-events]
  976 root      0.0  0.0 [kworker/u5:3-events_unbound]
  979 root      0.0  0.0 [kworker/1:1]
  980 root      0.0  0.0 [kworker/1:3]
  983 fathi     0.0  0.4 ps -eo pid,user,pcpu,pmem,cmd
```

Question 5

Building on the ps command used in Question 4, can you add an option to sort the listed processes by their memory usage (`pmem`)?

Command
`ps -eo pid,user,pcpu,pmem,cmd --sort=-pmem`
Output

```
fathi@secr2043:~$ ps -eo pid,user,pcpu,pmem,cmd --sort=-pmem
    PID USER     %CPU %MEM CMD
    358 root      0.0  2.7 /sbin/multipathd -d -s
    750 root      0.0  2.3 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
    669 root      0.0  1.3 /usr/libexec/udisks2/udisksd
      1 root      0.0  1.3 /sbin/init
    535 systemd+  0.0  1.3 /usr/lib/systemd/systemd-resolved
    766 root      0.0  1.3 /usr/sbin/ModemManager
    303 root      0.0  1.2 /usr/lib/systemd/systemd-journald
    519 systemd+  0.0  0.9 /usr/lib/systemd/systemd-networkd
    666 root      0.0  0.8 /usr/lib/systemd/systemd-logind
    910 root      0.0  0.8 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
    543 systemd+  0.0  0.7 /usr/lib/systemd/systemd-timesyncd
    651 polkitd   0.0  0.7 /usr/lib/polkit-1/polkitd --no-debug
    368 root      0.0  0.7 /usr/lib/systemd/systemd-udevd
    913 fathi     0.0  0.7 sshd: fathi@pts/0
    744 syslog    0.0  0.6 /usr/sbin/rsyslogd -n -iNONE
    645 message+  0.0  0.5 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
    911 root      0.0  0.5 sshd: fathi [priv]
    914 fathi     0.0  0.5 -bash
    984 fathi     0.0  0.4 ps -eo pid,user,pcpu,pmem,cmd --sort=-pmem
    836 root      0.0  0.2 /usr/sbin/cron -f -P
    844 root      0.0  0.1 /sbin/agetty -o -p -- \u --noclear - linux
    944 fathi     0.0  0.1 ./mainprocess
    947 fathi     0.0  0.1 ./subprocess1
    948 fathi     0.0  0.1 ./subprocess2
    949 fathi     0.0  0.1 ./subprocess1
    950 fathi     0.0  0.1 ./subprocess2
    951 fathi     0.0  0.1 ./subprocess2
    945 fathi     0.0  0.0 ./mainprocess
    946 fathi     0.0  0.0 ./mainprocess
```

Question 6

Construct a command using `ps`, suitable options, and any additional tools to visualize the hierarchical structure (tree-like) of the following processes:

- "mainprocess"
- "subprocess1"
- "subprocess2"

Command
`ps -ef --forest | grep -E "mainprocess|subprocess1|subprocess2"`
Output

```
fathi@secr2043:~$ ps -ef --forest | grep -E "mainprocess|subprocess1|subprocess2"
fathi        944     914  0 14:05 pts/0    00:00:00             \_ ./mainprocess
fathi        945     944  0 14:05 pts/0    00:00:00             |   \_ ./mainprocess
fathi        947     945  0 14:05 pts/0    00:00:00             |   |   \_ ./subprocess1
fathi        949     945  0 14:05 pts/0    00:00:00             |   |   \_ ./subprocess1
fathi        946     944  0 14:05 pts/0    00:00:00             |   \_ ./mainprocess
fathi        948     946  0 14:05 pts/0    00:00:00             |       \_ ./subprocess2
fathi        950     946  0 14:05 pts/0    00:00:00             |       \_ ./subprocess2
fathi        951     946  0 14:05 pts/0    00:00:00             |       \_ ./subprocess2
fathi       1024     914  0 14:40 pts/0    00:00:00             \_ grep --color=auto -E mainprocess|subprocess1|subprocess2
```

Question 7

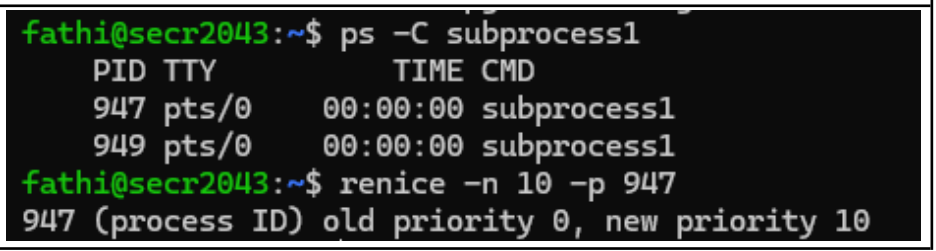
Use `pstree` command with option that show the number of threads to each process.

Command
`pstree -p -t`
Output

Question 8

Use `renice` command to change priority level of one of process “subprocess1”.

Command
`renice -n 10 -p 947`
Output



Question 9

Terminate all running processes with the name “mainprocess”.

Command
`pkill mainprocess`
Output

```
fathi@secr2043:~$ pkill mainprocess
Main process (ID: 945) received signal: 15. Terminating...
Main process (ID: 944) received signal: 15. Terminating...
Main process (ID: 946) received signal: 15. Terminating...
[1]+  Done                    ./mainprocess
```

Question 10

Write a short C or Python code (choose only one language) demonstrating multiprocessing with `fork()` and `wait()`. Compile and/or run the code. Show the output.

Source Code:

```
 GNU nano 7.2                          fork_example.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    } else if (pid == 0) {
        printf("Child process: PID = %d\n", getpid());
    } else {
        wait(NULL);
        printf("Parent process: PID = %d, Child PID = %d\n", getpid(), pid);
    }

    return 0;
}
```

Output:

```
fathi@secr2043:~/fork_example$ nano fork_example.c
fathi@secr2043:~/fork_example$ gcc fork_example.c -o fork_example
fathi@secr2043:~/fork_example$ ./fork_example
Child process: PID = 2563
Parent process: PID = 2562, Child PID = 2563
```